

Subject: Fundamentals of Engineering Mathematics -II**Module 1 : Differential Equations of First Order and First Degree****Exact Differential Equations****Introduction to Differential Equation of first order and first degree,
Exact Differential Equation**

●**Introduction:** Differential equations arise in many areas of science and technology, specifically whenever a deterministic relation involving some continuously varying quantities (modeled by function) and their rate of change in space and/or time (Expressed as derivatives) is known or postulated.

Examples:-

1. A stone is falling under gravity, if $y(t)$ is the position of the stone at the time t and g is the acceleration due to gravity then we have

$$\frac{d^2y}{dt^2} = g = \text{constant}$$

(we are not considering the air resistance) so finding the velocity as a function of time or displacement as a function of time involves solving a differential equation.

2. Take an ordinary spring that resist compression as well as extension and suspend it vertically from a fixed support. At the lower end of the spring we attach a body of mass m . We assume m to be so large that we may disregard the mass of the spring. If we pull the body down a certain distance and then release it, it undergoes a motion (assume that the body moves strictly vertically) the motion of this mechanical system is given by,

$$my'' + ky = 0$$

(y is acceleration of the body) and $y(t)$ is the displacement of the body at time t and k ($k > 0$) is called the spring constant (or spring module). So understanding the motion of the body involves solving a differential equation.

3. Kirchhoff's law: the algebraic sum of the voltage drop around any closed circuit is equal to resultant emf in the circuit.

Hence current flowing through L-R circuit can be expressed as the DE

$$Ri + L\frac{di}{dt} = E$$

and in C-R circuit as $R\frac{dq}{dt} + \frac{q}{c} = E$

4. Differential equations are used in studying motion of planets.

5. Differential equation is also used in studying **Resonance** phenomenon.

6. One popular application of differential equations (and in particular, first order linear differential equations) is in modeling the amount (or concentration) of a substance in a well-stirred tank/vessel subject to constant in- flow and out- flow. Common simple applications are

- an industrial mixing tank with an entry pipe (pumping the chemical of interest in) and an exit pipe
- a lake with an in flow (say, a river) feeding a pollutant from upstream and an out flow (also, a river) flowing downstream;
- a tub or sink with a steady in flow (say, a faucet) and a steady out flow (say, a drain).

So differential equation plays an important role in modeling physical, technical, or biological processes

●**Definition:** In Mathematics, a differential equation is an equation that contains one or more functions with its derivatives.

The derivatives of the function define the rate of change of a function at a point.

Examples:

1. $\frac{dy}{dx} + 4y = 0$Ordinary Differential Equation
2. $\frac{d^2y}{dx^2} = \cos x$Ordinary Differential Equation
3. $x^3 \frac{dy}{dx} + 2e^x \frac{d^2y}{dx^2} = 0$Ordinary Differential Equation
4. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$Partial Differential Equation

Note:

1. Ordinary differential equation (o.d.e) has one independent variable.
2. Partial Differential Equation (p.d.e) has more than one independent variables.

In this chapter we are going to see "Ordinary differential equation" only.

●**Order and Degree of DE:**

The **order** of an ordinary DE is the order of the highest derivative that appears in it

The **degree** of an ordinary DE is the exponent of the highest derivative appearing in a differential equation free from radical and fractions.

Examples:

1. $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + y = 6x$order=3
2. $\frac{d^2y}{dx^2} + e^{5y} = \log x$order=2
3. $\frac{d^3y}{dx^3} + \left(\frac{d^2y}{dx^2}\right)^5 + 6\frac{dy}{dx} = x$ order =3, degree=1
4. $\frac{d^2y}{dx^2} + e^{5y} = \log x$order=2, degree=1
5. $\left(\frac{dy}{dx}\right)^{2/3} = 7 + 2\frac{d^2y}{dx^2} \Rightarrow \left(\frac{dy}{dx}\right)^2 = \left(7 + 2\frac{d^2y}{dx^2}\right)^3$...order=2, degree=3

•Solutions of differential equations:-

A solution of DE is a functional relation between the independent variables and dependent variables involved such that this relation and derivative obtained from it satisfy the given DE.

For eg:- $y = 3x + 7$ is a solution of DE: $\frac{d^2y}{dx^2} = 0$ [one way of checking $y = 3x + 7$ is a solution is find the derivative and put it back in the equation.

$$y = 3x + 7$$

$$\frac{dy}{dx} = 3$$

$$\frac{d^2y}{dx^2} = 0$$

A DE has two types of solutions:

(1)**General solution (G.E.) of DE** A solution of a DE in which the number of arbitrary constants is equal to the order of the DE is called the General solution (G.S.) of the DE.

for eg:- $y = ax + b$, where a and b are arbitrary constants is the general solution of $\frac{d^2y}{dx^2} = 0$.

(2)**Particular equation of DE:-** A solution obtained from GS of DE by assigning particular values to arbitrary contents (by using initial conditions) is called as particular solution of DE. In this semester we are going to see linear ODE of order n (Linear means the power of the unknown function and its derivatives is 1 (products are not allowed))

We will first start with the first order first degree DE

Form: $\frac{dy}{dx} = f(x, y)$

or

$$M(x, y)dx + N(x, y)dy = 0$$

•Types of first order first degree DE

1. Variable-separable method
2. Homogeneous DE
3. Non-Homogeneous DE
4. **Exact DE**
5. **Linear DE**
6. **Bernoulli's DE**

Note: The method of solving a differential equation depends on its type.

We have already done the first 3 methods in the earlier classes.

Here you will solve differential equations of the 4th, 5th and 6th type.

4. Exact Differential Equations

The Differential Equation

$$M(x, y)dx + N(x, y)dy = 0$$

is called an Exact Differential Equation if there exist a function $U(x, y)$ such that

$$dU = Mdx + Ndy$$

That is a differential equation is called Exact if it is the result of a simple differentiation without any reduction or elimination.

Example: DE $x dx + y dy = 0$ is Exact

DE $y dx - x dy = 0$ is not Exact

Condition:- A necessary and sufficient condition for the DE

$Mdx + Ndy = 0$ to be exact is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Solution: The general solution of an exact DE is given by

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy = c \dots (*)$$

Note: If the partial derivatives of M and N are continuous then the above formula (*) takes one of the following forms.

$$\int M dx + \int (\text{terms of } N \text{ which are free from } x) dy = c \quad (1)$$

$$\int N dy + \int (\text{terms of } M \text{ which are free from } y) dx = c \quad (2)$$

Since the formula (*) is the general formula, it is used to solve any exact differential equation.

Examples: Solve the following differential equations:

1. $(\sec x \tan x \tan y - e^x)dx + (\sec x \sec^2 y)dy = 0$

Solution: The given DE is

$$(\sec x \tan x \tan y - e^x)dx + (\sec x \sec^2 y)dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned} M &= \sec x \tan x \tan y - e^x & \text{and} & & N &= \sec x \sec^2 y \\ \therefore \frac{\partial M}{\partial y} &= \sec x \tan x \sec^2 y & & & \frac{\partial N}{\partial x} &= \sec x \tan x \sec^2 y \\ & & \therefore \frac{\partial M}{\partial y} &= & \frac{\partial N}{\partial x} \end{aligned}$$

Hence the given DE is Exact.

Hence it's general solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy = c$$

$$\begin{aligned} \text{here, } \int M dx &= \int (\sec x \tan x \tan y - e^x) dx \\ &= \tan y \int \sec x \tan x dx - \int e^x dx \\ &= \tan y \sec x - e^x \end{aligned}$$

$$\begin{aligned} \therefore \frac{\partial}{\partial y} \int M dx &= \sec^2 y \sec x \\ \therefore N - \frac{\partial}{\partial y} \int M dx &= \sec x \sec^2 y - \sec^2 y \sec x \\ &= 0 \end{aligned}$$

$$\begin{aligned} \therefore \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy &= \int 0 dy \\ &= 0 \end{aligned}$$

Hence the solution is:

$$\tan y \sec x - e^x = c$$

2. $(\tan y + x)dx + (x \sec^2 y - 3y)dy = 0$

Solution: The given DE is

$$(\tan y + x)dx + (x \sec^2 y - 3y)dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned} M &= \tan y + x & \text{and} & & N &= x \sec^2 y - 3y \\ \therefore \frac{\partial M}{\partial y} &= \sec^2 y & & & \frac{\partial N}{\partial x} &= \sec^2 y \\ & & \therefore \frac{\partial M}{\partial y} &= & \frac{\partial N}{\partial x} \end{aligned}$$

Hence the given DE is Exact.

Hence it's general solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy = c$$

$$\begin{aligned} \text{here, } \int M dx &= \int (\tan y + x) dx \\ &= x \tan y + \frac{x^2}{2} \end{aligned}$$

$$\therefore \frac{\partial}{\partial y} \int M dx = x \sec^2 y$$

$$\begin{aligned} \therefore N - \frac{\partial}{\partial y} \int M dx &= x \sec^2 y - 3y - x \sec^2 y \\ &= -3y \end{aligned}$$

$$\begin{aligned} \therefore \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy &= \int -3y dy \\ &= -\frac{3y^2}{2} \end{aligned}$$

Hence the solution is:

$$\begin{aligned} x \tan y + \frac{x^2}{2} - \frac{3y^2}{2} &= c \\ \therefore 2x \tan y + x^2 - 3y^2 &= 2c \end{aligned}$$

3. $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$

Solution: The given DE is

$$\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$$

$$\text{i.e. } (y \cos x + \sin y + y)dx + (-\cos(xy) + xy \sin(xy) + \cos(xy) \\ nx + x \cos y + x)dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned} M &= y \cos x + \sin y + y & \text{and} & & N &= \sin x + x \cos y + x \\ \therefore \frac{\partial M}{\partial y} &= \cos x + \cos y + 1 & & & \frac{\partial N}{\partial x} &= \cos x + \cos y + 1 \\ & & \therefore \frac{\partial M}{\partial y} &= \frac{\partial N}{\partial x} \end{aligned}$$

Hence the given DE is Exact.

Hence it's general solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy = c$$

$$\begin{aligned} \text{here, } \int M dx &= \int (y \cos x + \sin y + y) dx \\ &= y \int \cos x dx + \sin y \int dx + y \int dx \\ &= y \sin x + x \sin y + yx \end{aligned}$$

$$\therefore \frac{\partial}{\partial y} \int M dx = \sin x + x \cos y + x$$

$$\begin{aligned} \therefore N - \frac{\partial}{\partial y} \int M dx &= (\sin x + x \cos y + x) - (\sin x + x \cos y + x) \\ &= 0 \end{aligned}$$

$$\therefore \int (N - \frac{\partial}{\partial y} \int M dx) dy = \int 0 dy = 0$$

Hence the solution is:

$$y \sin x + x \sin y + yx = c$$

4. $[y \sin(xy) + xy^2 \cos(xy)]dx + [x \sin(xy) + x^2y \cos(xy)]dy = 0$

Solution: The given DE is

$$[y \sin(xy) + xy^2 \cos(xy)]dx + [x \sin(xy) + x^2y \cos(xy)]dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$M = y \sin(xy) + xy^2 \cos(xy) \quad \text{and} \quad N = x \sin(xy) + x^2y \cos(xy)$$

$$\therefore \frac{\partial M}{\partial y} = \sin(xy) + y(\cos(xy))x + x[2y \cos(xy) - y^2(\sin(xy))x]$$

$$\therefore \frac{\partial M}{\partial y} = \sin(xy) + 3xy \cos(xy) - x^2y^2 \sin(xy)$$

$$\text{and, } \frac{\partial N}{\partial x} = [\sin(xy) + x(\cos(xy))y] + y[2x \cos(xy) - x^2(\sin(xy))y]$$

$$\therefore \frac{\partial N}{\partial x} = \sin(xy) + 3xy \cos(xy) - x^2y^2 \sin(xy)$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Hence the given DE is Exact.

Hence it's general solution is: $\int Mdx + \int (N - \frac{\partial}{\partial y} \int M dx) dy = c$

$$\text{here, } \int Mdx = \int (y \sin(xy) + xy^2 \cos(xy))dx$$

$$= y \frac{-\cos(xy)}{y} + y^2 \left[x \frac{\sin(xy)}{y} + 1 \frac{\cos(xy)}{y^2} \right]$$

$$= -\cos(xy) + xy \sin(xy) + \cos(xy)$$

$$= xy \sin(xy)$$

$$\therefore \frac{\partial}{\partial y} \int M dx = (\sin xy)x + x[y(\cos(xy))x]$$

$$= x \sin(xy) + x^2y \cos(xy) + x \sin(xy) - x \sin(xy)$$

$$= x \sin(xy) + x^2y \cos(xy)$$

$$\therefore N - \frac{\partial}{\partial y} \int M dx = (x \sin(xy) + x^2y \cos(xy)) - (x \sin(xy) + x^2y \cos(xy))$$

$$= 0$$

$$\therefore \int (N - \frac{\partial}{\partial y} \int M dx) dy = \int 0 dy = 0$$

Hence the solution is: $xy \sin(xy) = c$

Practice Problems:

1. Solve $(x^2 + 1)\frac{dy}{dx} = (x^3 - 2xy + x)$ (Ans : $-\frac{x^4}{4} + x^2y - \frac{x^2}{2} + y = c$)
2. Solve $(1 + \log(xy))dx + (1 + \frac{x}{y})dy = 0$ (Ans : $x \log y + x \log x + y = c$)
3. Solve $(x\sqrt{1 - x^2y^2} - y)dy + (x + y\sqrt{1 - x^2y^2})dx = 0$
4. $x dx + y dy = \frac{ax}{x^2 + y^2} dy - \frac{ay}{x^2 + y^2} dx$

Refresher Quiz

1. In an exact differential equation $Mdx + Ndy = 0$,

(a) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$

(b) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$

(c) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$

(d) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

2. An integrating factor makes a differential equation

First Order DE Reducible to Exact Form

Consider the equation $ydx - xdy = 0$(1)

on comparing with $Mdx + Ndy = 0$

we have, $M = y$ & $N = -x$

here

$$\frac{\partial M}{\partial y} = 1 \quad \text{and} \quad \frac{\partial N}{\partial x} = -1$$

so

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Hence equation is not exact. Now, question is: can we make the given equation as exact, If we multiply equation (1) by $\frac{1}{y^2}$ we get,

$$\frac{1}{y}dx - \frac{x}{y^2}dy = 0$$

Now new M and N are,

$$\begin{aligned} M &= \frac{1}{y} \quad \text{and} \quad N = \frac{-x}{y^2} \\ \frac{\partial M}{\partial y} &= -\frac{1}{y^2} \quad \frac{\partial N}{\partial x} = -\frac{1}{y^2} \\ \therefore \frac{\partial M}{\partial y} &= \frac{\partial N}{\partial x} \end{aligned}$$

So, Now equation has become exact and hence can be solved.

A factor which converts the given Non-exact DE into Exact DE is called as an Integrating factor (I.F.)

Note:-There may be more than one integrating factors for a given differential equation.

Rules for Finding I.F:

1. If $\frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N}$ is a function of 'x' only say $f(x)$ then

$$\text{I.F.} = e^{\int f(x)dx}$$

2. If $\frac{\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)}{M}$ is a function of 'y' only say $f(y)$ then

$$\text{I.F.} = e^{\int f(y)dy}$$

3. If the equation $M(x, y)dx + N(x, y)dy = 0$ is homogeneous DE and $Mx + Ny \neq 0$ then

$$\text{I.F.} = \frac{1}{Mx + Ny}$$

4. If equation is of the form $f(xy)ydx + g(xy)x dy = 0$ and $Mx - Ny \neq 0$ then

$$\text{I.F.} = \frac{1}{Mx - Ny}$$

Examples: Solve following Differential Equations

Rule 1: If $\frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N}$ is a function of 'x' only say $f(x)$ then

$$\text{I.F.} = e^{\int f(x)dx}$$

1. Solve $(2x \log x - xy)dy + 2ydx = 0$

Solution: The given DE is

$$2ydx + (2x \log x - xy)dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned} M &= 2y & \text{and} & & N &= 2x \log x - xy \\ \therefore \frac{\partial M}{\partial y} &= 2 & & & \frac{\partial N}{\partial x} &= 2(1 + \log x) - y \\ & & & & \therefore \frac{\partial M}{\partial y} &\neq \frac{\partial N}{\partial x} \end{aligned}$$

Hence the given DE is not exact.

Now, $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2 - [2(1 + \log x) - y] = y - 2 \log x$

$$\begin{aligned} \therefore \frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N} &= \frac{2 - [2(1 + \log x) - y]}{2x \log x - xy} \\ &= \frac{y - 2 \log x}{2x \log x - xy} \\ &= \frac{y - 2 \log x}{-x(y - 2 \log x)} \\ &= \frac{-1}{x} \\ &= f(x) \text{ (function of } x) \end{aligned}$$

$$\therefore I.F. = e^{\int f(x) dx} = e^{\frac{-1}{x}} = e^{-\log x} = \frac{1}{x}$$

Multiplying the given DE by this I.F. we have,

$$\underbrace{\frac{2y}{x}}_M dx + \underbrace{(2 \log x - y)}_N dy = 0$$

which is exact.

Hence it's general solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx\right) dy = c$$

$$\begin{aligned} \text{here, } \int M dx &= \int \frac{2y}{x} dx \\ &= 2y \log x \end{aligned}$$

$$\begin{aligned} \therefore \int \left(N - \frac{\partial}{\partial y} \int M dx\right) dy &= \int N (\text{terms in } N \text{ free from } x) dy \\ &= \int -y dy \\ &= -\frac{y^2}{2} \end{aligned}$$

Hence the solution is:

$$2 \log x - \frac{y^2}{2} = c$$

2. $(xy^2 - e^{1/x^3})dx - x^2ydy = 0$

Solution: The given DE is

$$(xy^2 - e^{1/x^3})dx - x^2ydy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned} M &= xy^2 - e^{1/x^3} & \text{and} & \quad N = -x^2y \\ \therefore \frac{\partial M}{\partial y} &= 2xy & \quad \frac{\partial N}{\partial x} &= -2xy \\ & & \therefore \frac{\partial M}{\partial y} &\neq \frac{\partial N}{\partial x} \end{aligned}$$

Hence the given DE is not exact.

Now,

$$\begin{aligned} \therefore \frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N} &= \frac{2xy - (-2xy)}{-x^2y} \\ &= \frac{4xy}{-x^2y} \\ &= \frac{-4}{x} \\ &= f(x) \text{ (function of } x) \end{aligned}$$

$$\therefore I.F. = e^{\int f(x)dx} = e^{\int \frac{-4}{x} dx} = e^{-4 \log x} = \frac{1}{x^4}$$

Multiplying the given DE by this I.F. we have,

$$\begin{aligned} \left(\frac{xy^2 - e^{1/x^3}}{x^4}\right)dx - \frac{x^2y}{x^4}dy &= 0 \\ \therefore \underbrace{\left(\frac{y^2}{x^3} - \frac{e^{1/x^3}}{x^4}\right)}_M dx - \underbrace{\frac{y}{x^2}}_N dy &= 0 \end{aligned}$$

which is exact.

The general solution is given by

$$\int Mdx + \int \left(N - \frac{\partial}{\partial y} \int M dx\right) dy = c$$

$$\begin{aligned}
\text{here, } \int M dx &= \int \left(\frac{y^2}{x^3} - \frac{e^{1/x^3}}{x^4} \right) dx \\
&= y^2 \int \frac{1}{x^3} dx - \int \frac{e^{1/x^3}}{x^4} dx \\
&= y^2 \frac{x^{-3+1}}{-3+1} - \int e^t \left(\frac{-dt}{3} \right) \\
&= \frac{-y^2 x^{-2}}{2} + \frac{1}{3} e^{1/x^3} \\
&= \frac{e^{1/x^3}}{3} - \frac{y^2}{2x^2}
\end{aligned}$$

$$\begin{aligned}
\therefore \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy &= \int N(\text{terms in N free from x}) dy \\
&= \int 0 dy \\
&= 0
\end{aligned}$$

Hence the solution is:

$$\frac{e^{1/x^3}}{3} - \frac{y^2}{2x^2} = c$$

$$3. \left(y + \frac{y^3}{3} + \frac{x^2}{2} \right) dx + \frac{1}{4}(x + xy^2) dy = 0$$

Solution: The given DE is

$$\left(y + \frac{y^3}{3} + \frac{x^2}{2} \right) dx + \frac{1}{4}(x + xy^2) dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned}
M &= y + \frac{y^3}{3} + \frac{x^2}{2} & \text{and} & & N &= \frac{1}{4}(x + xy^2) \\
\therefore \frac{\partial M}{\partial y} &= 1 + y^2 & & & \frac{\partial N}{\partial x} &= \frac{1}{4}(1 + y^2) \\
& & & & \therefore \frac{\partial M}{\partial y} &\neq \frac{\partial N}{\partial x}
\end{aligned}$$

Hence the given DE is not exact.

Now,

$$\begin{aligned}\therefore \frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N} &= \frac{(1 + y^2) - \frac{1}{4}(1 + y^2)}{\frac{1}{4}(x + xy^2)} \\ &= \frac{\frac{3}{4}(1 + y^2)}{\frac{x}{4}(1 + y^2)} \\ &= \frac{3}{x} \\ &= f(x) \text{ (function of } x \text{)}\end{aligned}$$

$$\therefore I.F. = e^{\int f(x)dx} = e^{\int \frac{3}{x} dx} = e^{3 \log x} = x^3$$

Multiplying the given DE by this I.F. we have,

$$\underbrace{\left(x^3y + \frac{x^3y^3}{3} + \frac{x^5}{2}\right) dx}_{M} + \underbrace{\frac{1}{4}(x^4 + x^4y^2) dy}_N = 0$$

which is exact.

Hence it's general solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx\right) dy = c$$

$$\begin{aligned}\text{here, } \int M dx &= \int \left(x^3y + \frac{x^3y^3}{3} + \frac{x^5}{2}\right) dx \\ &= y \frac{x^4}{4} + \frac{y^3}{3} \frac{x^4}{4} + \frac{x^6}{12} \\ &= \frac{yx^4}{4} + \frac{y^3x^4}{12} + \frac{x^6}{12}\end{aligned}$$

$$\begin{aligned}\therefore \int \left(N - \frac{\partial}{\partial y} \int M dx\right) dy &= \int N(\text{terms in } N \text{ free from } x) dy \\ &= \int 0 dy \\ &= 0\end{aligned}$$

Hence the solution is:

$$\begin{aligned}\frac{yx^4}{4} + \frac{y^3x^4}{12} + \frac{x^6}{12} &= c \\ \therefore 3yx^4 + y^3x^4 + x^6 &= c\end{aligned}$$

Rule 2: If $\frac{\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)}{M}$ is a function of 'y' only say $f(y)$ then

$$\text{I.F.} = e^{\int f(y) dy}$$

$$1. (2xy^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0$$

Solution: The given DE is

$$(2xy^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned} M &= 2xy^4e^y + 2xy^3 + y & \text{and} & & N &= x^2y^4e^y - x^2y^2 - 3x \\ \therefore \frac{\partial M}{\partial y} &= 2x(4y^3e^y + y^4e^y) + 6x^2 + 1 & & & \frac{\partial N}{\partial x} &= 2xy^4e^y - 2xy^2 - 3 \\ & & \therefore \frac{\partial M}{\partial y} &\neq \frac{\partial N}{\partial x} \end{aligned}$$

Hence the given DE is not exact.

Now,

$$\begin{aligned} \therefore \frac{\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)}{M} &= \frac{(2xy^4e^y - 2xy^2 - 3) - (2x(4y^3e^y + y^4e^y) + 6x^2 + 1)}{2xy^4e^y + 2xy^3 + y} \\ &= \frac{(2xy^4e^y - 2xy^2 - 3) - (8xy^3e^y + 2xy^4e^y + 6x^2 + 1)}{2xy^4e^y + 2xy^3 + y} \\ &= \frac{-2xy^2 - 3 - 8xy^3e^y - 6xy^2 - 1}{2xy^4e^y + 2xy^3 + y} \\ &= \frac{-8xy^3e^y - 8xy^2 - 4}{2xy^4e^y + 2xy^3 + y} \\ &= \frac{-4}{y} \\ &= f(y) \text{ (function of y)} \end{aligned}$$

$$\therefore \text{I.F.} = e^{\int f(y) dy} = e^{\int \frac{-4}{y} dy} = e^{-4 \log y} = \frac{1}{y^4}$$

Multiplying the given DE by this I.F. we have,

$$\underbrace{\left(2xe^y + \frac{2x}{y} + \frac{1}{y^3}\right)}_M dx + \underbrace{\left(x^2e^y - \frac{x^2}{y^2} - \frac{-3x}{y^4}\right)}_N dy = 0$$

which is exact.

Hence it's general solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy = c$$

$$\begin{aligned} \text{here, } \int M dx &= \int \left(2xe^y + \frac{2x}{y} + \frac{1}{y^3} \right) dx \\ &= x^2 e^y + \frac{x^2}{y} + \frac{x}{y^3} \end{aligned}$$

$$\begin{aligned} \therefore \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy &= \int N(\text{terms in N free from x}) dy \\ &= \int 0 dy \\ &= 0 \end{aligned}$$

Hence the solution is:

$$x^2 e^y + \frac{x^2}{y} + \frac{x}{y^3} = c$$

2. $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$

Solution: The given DE is

$$(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned} M &= y^4 + 2y & \text{and} & & N &= xy^3 + 2y^4 - 4x \\ \therefore \frac{\partial M}{\partial y} &= 4y^3 + 2 & & & \frac{\partial N}{\partial x} &= y^3 - 4 \\ & & & & \therefore \frac{\partial M}{\partial y} &\neq \frac{\partial N}{\partial x} \end{aligned}$$

Hence the given DE is not exact.

Now,

$$\begin{aligned}
\therefore \frac{\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)}{M} &= \frac{(y^3 - 4) - (4y^3 + 2)}{y^4 + 2y} \\
&= \frac{-3y^3 - 6}{y^4 + 2y} \\
&= \frac{-3(y^3 + 2)}{y(y^3 + 2)} \\
&= \frac{-3}{y} \\
&= f(y) \text{ (function of } y)
\end{aligned}$$

$$\therefore I.F. = e^{\int f(y)dy} = e^{\int \frac{-3}{y} dy} = e^{-3 \log y} = \frac{1}{y^3}$$

Multiplying the given DE by this I.F. we have,

$$\begin{aligned}
\left(\frac{y^4 + 2y}{y^3}\right) dx + \left(\frac{xy^3 + 2y^4 - 4x}{y^3}\right) dy &= 0 \\
\underbrace{\left(y + \frac{2}{y^2}\right) dx}_M + \underbrace{\left(x + 2y - 4\frac{x}{y^3}\right) dy}_N &= 0
\end{aligned}$$

which is exact.

Hence it's general solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx\right) dy = c$$

$$\begin{aligned}
\text{here, } \int M dx &= \int \left(y + \frac{2}{y^2}\right) dx \\
&= \left(y + \frac{2}{y^2}\right) x \\
&= xy + \frac{2x}{y^2}
\end{aligned}$$

$$\begin{aligned}
\therefore \int \left(N - \frac{\partial}{\partial y} \int M dx\right) dy &= \int N(\text{terms in } N \text{ free from } x) dy \\
&= \int 2y dy \\
&= y^2
\end{aligned}$$

Hence the solution is:

$$xy + \frac{2x}{y^2} + y^2 = c$$

Refresher Quiz

1. The integrating factor of the DE $xydx + (x^2 - xy)dy = 0$, is

(a) $\frac{-1}{x}$ (b) $\frac{-1}{x^2}$

(c) $\frac{1}{x}$ (d) $\frac{1}{x^2}$

2. $\int x^2 e^{-x}$ equals

(a) $-e^{-x}(x^2 + 2x + 2)$ (b) $e^{-x}(x^2 + 2x + 2)$

(c) $e^{-x}(x^2 - 2x + 2)$ (d) $e^{-x}(-x^2 + 2x - 2)$

First Order DE Reducible to Exact Form

Rule 3: If the equation $M(x, y)dx + N(x, y)dy = 0$ is homogeneous DE and $Mx + Ny \neq 0$ then I.F. = $\frac{1}{Mx + Ny}$

1. DE $Mdx + Ndy = 0$ is said to be Homogeneous DE if M & N are Homogeneous functions of same degree.
2. Recall from the 1st semester a function $f(x, y)$ is said to be a homogeneous function of degree m if $f(\lambda x, \lambda y) = \lambda^m f(x, y)$

1. Solve $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$

Solution: The given DE is

$$(x^2y - 2xy^2)dx + (3x^2y - x^3)dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned} M &= (x^2y - 2xy^2) & \text{and} & & N &= (3x^2y - x^3) \\ \therefore \frac{\partial M}{\partial y} &= x^2 - 4xy & & & \frac{\partial N}{\partial x} &= 6xy - 3x^2 \\ & & \therefore \frac{\partial M}{\partial y} &\neq \frac{\partial N}{\partial x} \end{aligned}$$

Hence the given DE is not exact.

Now,

$$M = (x^2y - 2xy^2) \quad \text{and} \quad N = (3x^2y - x^3)$$

Both are Homogeneous functions of degree 3 and

$$\begin{aligned} Mx + Ny &= (x^2y - 2xy^2)x + (3x^2y - x^3)y \\ &= x^3y - 2x^2y^2 + 3x^2y^2 - x^3y \\ &= x^2y^2 \neq 0 \end{aligned}$$

$$\therefore I.F. = \frac{1}{Mx + Ny} = \frac{1}{x^2y^2}$$

Multiplying the given DE by this I.F. we have,

$$\frac{(x^2y - 2xy^2)}{x^2y^2}dx + \frac{(3x^2y - x^3)}{x^2y^2}dy = 0$$

$$\underbrace{\left(\frac{1}{y} - \frac{2}{x}\right)}_M dx + \underbrace{\left(\frac{3}{y} - \frac{x}{y^2}\right)}_N dy = 0$$

which is exact.

Hence it's general solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx\right) dy = c$$

$$\begin{aligned} \text{here, } \int M dx &= \int \left(\frac{1}{y} - \frac{2}{x}\right) dx \\ &= \frac{x}{y} - 2 \log x \end{aligned}$$

$$\begin{aligned} \therefore \int \left(N - \frac{\partial}{\partial y} \int M dx\right) dy &= \int N(\text{terms in N free from x}) dy \\ &= \int \frac{3}{y} dy \\ &= 3 \log y \end{aligned}$$

Hence the solution is:

$$\frac{x}{y} - 2 \log x + 3 \log y = c$$

2. Solve $(y(x + y))dx - (x(y - x))dy = 0$

Solution: The given DE is

$$(y(x + y))dx + (x(x - y))dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned} M &= xy + y^2 & \text{and} & & N &= x^2 - xy \\ \therefore \frac{\partial M}{\partial y} &= x + 2y & & & \frac{\partial N}{\partial x} &= 2x - y \\ & & \therefore \frac{\partial M}{\partial y} &\neq \frac{\partial N}{\partial x} \end{aligned}$$

Hence the given DE is not exact.

Now,

$$M = xy + y^2 \quad \text{and} \quad N = x^2 - xy$$

Both are Homogeneous functions of degree 2 and

$$\begin{aligned} Mx + Ny &= (xy + y^2)x + (x^2 - xy)y \\ &= x^2y + xy^2 + x^2y - xy^2 \\ &= 2x^2y \neq 0 \end{aligned}$$

$$\therefore I.F. = \frac{1}{Mx + Ny} = \frac{1}{2x^2y}$$

Multiplying the given DE by this I.F. we have,

$$\begin{aligned} \frac{(xy + y^2)}{2x^2y} dx + \frac{(x^2 - xy)}{2x^2y} dy &= 0 \\ \underbrace{\left(\frac{1}{2x} + \frac{y}{2x^2} \right)}_M dx + \underbrace{\left(\frac{1}{2y} - \frac{1}{2x} \right)}_N dy &= 0 \end{aligned}$$

which is exact.

Hence it's general solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy = c$$

$$\begin{aligned}\text{here, } \int M dx &= \int \left(\frac{1}{2x} - \frac{y}{2x^2} \right) dx \\ &= \frac{1}{2} \log x - \frac{y}{2x}\end{aligned}$$

$$\begin{aligned}\therefore \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy &= \int N(\text{terms in N free from x}) dy \\ &= \int \frac{1}{2y} dy \\ &= \frac{1}{2} \log y\end{aligned}$$

Hence the solution is:

$$\begin{aligned}\frac{1}{2} \log x - \frac{y}{2x} + \frac{1}{2} \log y &= c \\ \frac{1}{2} \log(xy) - \frac{y}{2x} &= c\end{aligned}$$

Rule 4: If equation is of the form **$f(xy)ydx + g(xy)x dy = 0$ and $Mx - Ny \neq 0$ then**

$$\text{I.F.} = \frac{1}{Mx - Ny}$$

$$1. \text{ Solve } (y - xy^2)dx - (x + x^2y)dy = 0$$

Solution: The given DE is

$$(y - xy^2)dx + (-x - x^2y)dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned} M &= (y - xy^2) & \text{and} & & N &= (-x - x^2y) \\ \therefore \frac{\partial M}{\partial y} &= 1 - 2xy & & & \frac{\partial N}{\partial x} &= -1 - 2xy \\ & & \therefore \frac{\partial M}{\partial y} &\neq \frac{\partial N}{\partial x} \end{aligned}$$

Hence the given DE is not exact.

Now, The given DE can be written as

$$(1 - xy)ydx + (-1 - xy)x dy = 0$$

So it is in the form $f(xy)ydx + g(xy)x dy = 0$

and

$$\begin{aligned} Mx - Ny &= [(1 - xy)y]x - [(-1 - xy)x]y \\ &= xy - x^2y^2 + xy + x^2y^2 \\ &= 2xy \neq 0 \end{aligned}$$

$$\therefore \text{I.F.} = \frac{1}{Mx - Ny} = \frac{1}{2xy}$$

Multiplying the given DE by this I.F. we have,

$$\frac{(y - xy^2)}{2xy}dx + \frac{(-x - x^2y)}{2xy}dy = 0 \underbrace{\left(\frac{1}{2x} - \frac{y}{2}\right)}_M dx + \underbrace{\left(-\frac{1}{2y} - \frac{x}{2}\right)}_N dy = 0$$

which is exact.

Hence it's general solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx\right) dy = c$$

$$\begin{aligned}\text{here, } \int M dx &= \int \left(\frac{1}{2x} - \frac{y}{2} \right) dx \\ &= \frac{1}{2} \log x - \frac{xy}{2}\end{aligned}$$

$$\begin{aligned}\therefore \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy &= \int N (\text{terms in } N \text{ free from } x) dy \\ &= \int -\frac{1}{2y} dy \\ &= -\frac{1}{2} \log y\end{aligned}$$

Hence the solution is:

$$\begin{aligned}\frac{1}{2} \log x - \frac{xy}{2} - \frac{1}{2} \log y &= c \\ \therefore \log x - \log y - xy &= 2c \\ \log \frac{x}{y} - xy &= 2c\end{aligned}$$

2. Solve $(xy \sin y + \cos xy)ydx + (xy \sin y - \cos xy)x dy = 0$

Solution: The given DE is

$$(xy \sin y + \cos xy)ydx + (xy \sin y - \cos xy)x dy = 0$$

On comparing the given DE with $Mdx + Ndy = 0$ we get,

$$\begin{aligned}M &= (xy \sin y + \cos xy)y & \text{and} & & N &= (xy \sin y - \cos xy)x \\ \therefore M &= (xy^2 \sin y + y \cos xy) & \text{and} & & N &= (x^2 y \sin y - x \cos xy) \\ \frac{\partial M}{\partial y} &= x(2y \sin y + y^2 \cos y) + (\cos(xy) - y \sin(xy)x) \\ \therefore \frac{\partial M}{\partial y} &= 2xy \sin y + xy^2 \cos y + x \cos(xy) - xy \sin(xy) \\ & & \text{and} & & \\ \frac{\partial N}{\partial x} &= 2xy \sin y - y(\cos(xy) - x \sin(xy)y) \\ \therefore \frac{\partial N}{\partial x} &= 2xy \sin y - y \cos(xy) + xy^2 \sin(xy) \\ & & \therefore \frac{\partial M}{\partial y} &\neq \frac{\partial N}{\partial x}\end{aligned}$$

Hence the given DE is not exact.

Now, The given DE is in the form $f(xy)ydx + g(xy)xdy = 0$
and

$$\begin{aligned} Mx - Ny &= [xy^2 \sin y + y \cos(xy)]x - [x^2y \sin y - x \cos xy]y \\ &= x^2y^2 \sin y + xy \cos(xy) - [x^2y^2 \sin y - xy \cos xy] \\ &= 2xy \cos(xy) \neq 0 \end{aligned}$$

$$\therefore I.F. = \frac{1}{Mx - Ny} = \frac{1}{2xy \cos(xy)}$$

Multiplying the given DE by this I.F. we have,

$$\frac{(xy^2 \sin y + y \cos(xy))}{2xy \cos(xy)} dx - \frac{(x^2y \sin y - x \cos xy)}{2xy \cos(xy)} dy = 0$$

$$\underbrace{\left(\frac{y \tan(xy)}{2} + \frac{1}{2x} \right)}_M dx + \underbrace{\left(\frac{x \tan(xy)}{2} - \frac{1}{2y} \right)}_N dy = 0$$

which is exact.

Hence it's general solution is:

$$\int M dx + \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy = c$$

$$\begin{aligned} \text{here, } \int M dx &= \int \left(\frac{y \tan(xy)}{2} + \frac{1}{2x} \right) dx \\ &= \left(\frac{y}{2} \int \tan(xy) dx + \frac{1}{2} \int \frac{1}{x} dx \right) \\ &= \frac{y \log[\sec(xy)]}{2} + \frac{\log x}{2} \\ &= \frac{\log[\sec(xy)]}{2} + \frac{\log x}{2} \end{aligned}$$

$$\begin{aligned} \therefore \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy &= \int N(\text{terms in N free from x}) dy \\ &= \int -\frac{1}{2y} dy \\ &= -\frac{1}{2} \log y \end{aligned}$$

Hence the solution is:

$$\frac{\log[\sec(xy)]}{2} + \frac{\log x}{2} - \frac{1}{2} \log y = c$$

$$\therefore \log[\sec(xy) + \log \frac{x}{y}] = 2c$$

3. If x^n is an I.F. of $(x^4e^x - 2mxy^2)dx + 2mx^2ydy = 0$. find n and solve the equation.

Solution: On multiplying the given DE by IF we get,

$$(x^4e^x - 2mxy^2)x^n dx + 2mx^{n+2}ydy = 0 \text{---(1)}$$

Now this equation has become exact.

Therefore

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}[x^{n+4}e^x - 2mx^{n+1}y^2] = \frac{\partial}{\partial x}[2mx^{n+2}y]$$

$$0 - (2mx^{n+1})(2y) = 2m(n+2)yx^{n+1}$$

$$\therefore -2 = n + 2$$

$$\therefore n = -4$$

Putting $n = -4$ in equation (1) we have

$$(x^4e^x - 2mxy^2)x^{-4}dx + 2mx^{-2}ydy = 0$$

$$\text{i.e. } (e^x - \frac{2my^2}{x^3})dx + \frac{2my}{x^2}dy = 0$$

Hence it's general solution is:

$$\int Mdx + \int (N - \frac{\partial}{\partial y} \int M dx) dy = c$$

$$\begin{aligned}\text{here, } \int M dx &= \int \left(e^x - \frac{2my^2}{x^3} \right) dx \\ &= e^x + \frac{my^2}{x^2}\end{aligned}$$

$$\begin{aligned}\therefore \int \left(N - \frac{\partial}{\partial y} \int M dx \right) dy &= \int N(\text{terms in N free from x}) dy \\ &= \int 0 dy \\ &= 0\end{aligned}$$

Hence the solution is:

$$e^x + \frac{my^2}{x^2} = c$$

Practice Problems:

1. Solve $(y - 2x^3)dx - x(1 - xy)dy = 0$
2. Solve $(2x \log x - xy)dy + 2ydx = 0$
3. $(x^4 + y^4)dx - xy^3dy = 0$
4. Solve $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$
5. Solve $(x^4 + y^4)dx - xy^3dy = 0$
—(Ans: I.F = $\frac{1}{x^5}$..G.S : $y^4 = 4x^4 \log x + c'x^4$)
6. Solve $\frac{dy}{dx} + \frac{x^2y^3 + 2y}{2x - 2x^3y^2} = 0$
7. If $(x + y + 1)^n$ is an integrating factor of the equation $(2xy - y^2 - y)dx + (2xy - x^2 - x)dy = 0$. Find n and solve the equation.
(Ans: $n=-4$, Solution: $(x + y + 1)^3 = c(xy)$)
8. If y^n is an integrating factor of the equation $y(2x^2y + e^x)dx - (e^x + y^2)dy = 0$. Find n and solve the equation.
(Ans: $n=-2$, Solution: $\frac{2x^3}{3} + \frac{e^x}{y} - y = c$)

Refresher Quiz:

1. $\int \frac{1}{x \log x} dx$ equals

- (a) $\log x$ (b) $\log(\log x)$
 (c) $\log \frac{1}{x}$ (d) $\log(\log \frac{1}{x})$

2. $\int \sqrt{a^2 - x^2} dx$ equals

- (a) $\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$ (b) $\frac{x}{2} \sqrt{a^2 - x^2} - \frac{a^2}{2} \sin^{-1} \frac{x}{a}$
 (c) $\frac{a}{2} \sqrt{a^2 - x^2} + \frac{x}{2} \sin^{-1} \frac{x}{a}$ (d) $\frac{x^2}{2} \sqrt{a^2 - x^2} + \frac{a}{2} \sin^{-1} \frac{x}{a}$

Rules for Finding I.F:

1. If $\frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N}$ is a function of 'x' only say $f(x)$ then

$$\text{I.F.} = e^{\int f(x) dx}$$

2. If $\frac{\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)}{M}$ is a function of 'y' only say $f(y)$ then

$$\text{I.F.} = e^{\int f(y) dy}$$

3. If the equation $M(x, y)dx + N(x, y)dy = 0$ is a homogeneous DE and $Mx + Ny \neq 0$ then

$$\text{I.F.} = \frac{1}{Mx + Ny}$$

4. If equation is of the form $f(xy)ydx + g(xy)xdy = 0$ and $Mx - Ny \neq 0$ then

$$\text{I.F.} = \frac{1}{Mx - Ny}$$